

Introduction to dynamical systems

- Phase space : coordinates to describe "dynamic" variables
- Map
- Flow

Law of evolution on the phase space : dynamical system

$$\frac{dx}{dt} = v(t, x)$$

$x \in \text{Phase space}$
 $\mathbb{R}^d$

v : vector field

$$v(t, \underset{\substack{\uparrow \\ \text{base point}}}{x}) \in \mathbb{R}^d$$

$v(t, x) \in \text{tangent space at point } x$

"velocity" vector that describes the infinitesimal perturbation at point x at time t .

Autonomous dynamical system : v is independent of time

Non-autonomous dynamics : v depends on time.

Implicitly :

$$F\left(x, \frac{dx}{dt}, \frac{d^2x}{dt^2}, \dots, \frac{d^kx}{dt^k}\right) = 0$$

↓ convert into first order system

$$x_0 = x$$

$$x_1 = \frac{dx}{dt}$$

$$x_2 = \frac{d^2x}{dt^2}$$

⋮

$$x_{k-1} = \frac{d^{k-1}x}{dt^{k-1}}$$

$$z = \begin{bmatrix} x_0 \\ \vdots \\ x_{k-1} \end{bmatrix}$$

$$\frac{dz}{dt} = v(t, z)$$

$$x \in \mathbb{R}^d$$

$$z \in \mathbb{R}^{dk}$$

WLOG, we look at first order ODEs:

$$\frac{dx}{dt} = v(t, x)$$

$x \in \mathbb{R}^d$ phase space.

Continuous-time dynamical system : Flow.

ODEs

Dynamics

Initial-value problem

All possible solutions of the ODE

$$\left. \begin{array}{l} \frac{dx}{dt} = v(t, x) \\ x(0) = a \end{array} \right\}$$

Geometry in phase space that solutions occupy

Numerical time integration

Existence?

Control?

Uniqueness?

Regularity?

Volume of initial conditions → how does this evolve?
→ Ergodic theory
Relationship b/w long-time evolution of a single solution & volume in phase space

Perturbations!

Solution :

Trajectory / orbit of the dynamics

$$\frac{dx}{dt} = v(t, x) \quad x(0) = a$$

Dynamics deals with a qualitative understanding of the orbit structure

$$x(t) = \varphi^t(x(0))$$

↓
Flow

$$\text{Group: } \varphi^{t+s}(x(0)) = \varphi^t(\varphi^s(x(0)))$$

$$\varphi^{t+s} = \varphi^t \circ \varphi^s$$

↑

function composition

$$= \varphi^s \cdot \varphi^t$$

holds for autonomous dynamics

Discrete-time dynamics : map

$$\text{For example } \frac{dx}{dt} = v(t, x)$$

$$x(0), x(\tau), x(2\tau), \dots$$

$$x(\tau) = \varphi^\tau(x(0))$$

"

$$x_{t+1} = F_t x_t \quad t \in \mathbb{Z}$$

Autonomous dynamics

$$x_{t+1} = F(x_t)$$

$$F: \mathbb{R}^d \rightarrow \mathbb{R}^d$$

Self-map

$$x_t = F^t(x_0)$$

$$= \underbrace{F \circ \dots \circ F}_{t \text{ times}}(x_0)$$

Examples

- Low-dimensional dynamics
- Climate dynamics: atmosphere, ocean, vegetation, people, ...
- Chaos: sensitive dependence on initial conditions
Poincaré

$$\frac{d}{dt} x = \underline{v}(t, x)$$

$$x \in \mathbb{R}^d$$

$$d \approx 10^{10}$$

Lorenz equations: $\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix}$

modeling atmospheric convection

- $\frac{dx}{dt} = v(t, x)$

$$x \in \mathbb{R}^3$$

$$v(t, x)$$

passive scalar in a fluid



passive scalar turbulence

$$\Pr(x_{10^6}(t) > 200^\circ\text{C})?$$

Computational dynamics: approximating answers to questions in perturbation & ergodic theory

Linear dynamics

$$\frac{dx}{dt} = Ax.$$

$$A \in \mathbb{R}^{d \times d}.$$

$$x(0)$$

$$x(t) ?$$

$$x(t) = e^{tA} x(0).$$

Case 1: A is diagonalizable

$$P^{-1}AP = \Lambda$$

$\downarrow$ $\hookrightarrow$ diagonal

P : matrix with eigenvectors of A in its columns.